

MECHATRONICS AND IOT

ASSIGNMENT – 12

RFID-Based Smart Identification and IoT Monitoring System

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RFID-Based Smart Identification and IoT Monitoring System

1. Project Overview

The RFID-Based Smart Identification and IoT Monitoring System is an IoT-based identification system developed using an ESP8266 NodeMCU, RC522 RFID reader, RFID cards/tags, and an I2C LCD display. The system identifies a person by reading the unique ID (UID) stored in an RFID card.

The ESP8266 receives the UID from the RC522 RFID reader, compares it with the registered UID stored in the program, and identifies the corresponding person by name. The person's name is displayed on the LCD. At the same time, the identification data is transmitted through Wi-Fi to Adafruit IO, where it can be monitored using feeds and dashboards.

The system can be used for attendance monitoring, access identification, employee/student identification, and smart entry monitoring.

2. Project Statement

Traditional identification and attendance systems often require manual entry, which can be time-consuming and prone to errors. There is a need for a simple, low-cost and automated system that can identify registered users quickly and store their identification information for remote monitoring.

Therefore, this project aims to develop an RFID-based identification system with IoT connectivity that can automatically identify a registered person, display their name locally, and transmit the information to an online platform.

3. Proposed Solution

The proposed system uses RFID technology for contactless identification.

When an RFID card is placed near the RC522 RFID reader, the reader obtains the card's unique identification number. The ESP8266 processes this UID and compares it with the registered UID database programmed into the system.

If the UID matches a registered user:

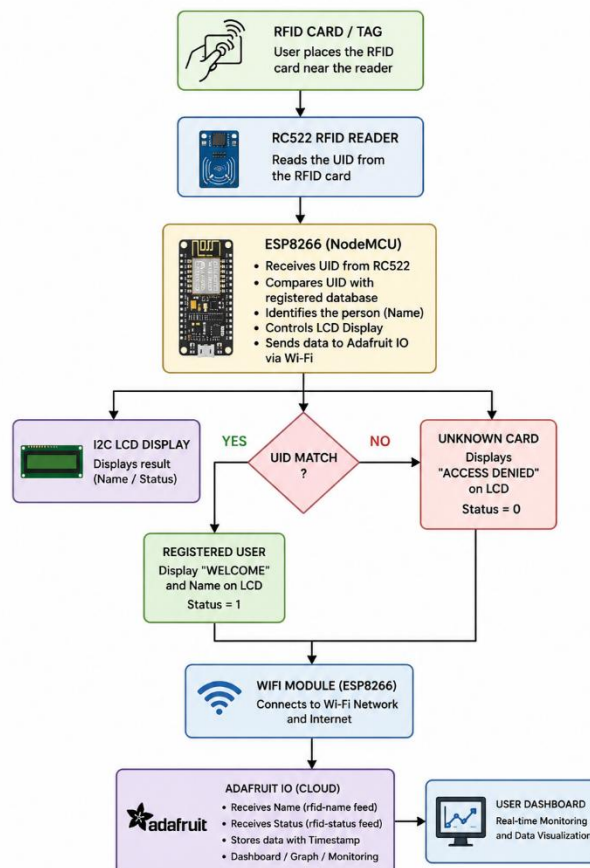
- The person's name is identified.
- The name is displayed on the LCD.
- The identification data is sent to Adafruit IO.
- The RFID status is recorded for monitoring.

If the UID does not match:

- The LCD displays "UNKNOWN CARD".
- The system displays "ACCESS DENIED".
- The event can be recorded as an unsuccessful identification.

4. System Architecture:

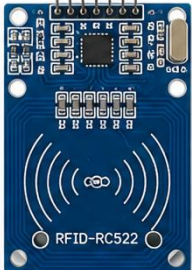
SYSTEM ARCHITECTURE



5.Circuit Table:

CIRCUIT TABLE

1. RC522 RFID READER TO ESP8266 (NODEMCU) CONNECTIONS

RC522 RFID Reader 	RC522 Pin	ESP8266 (NodeMCU) Pin	Function	Description
	3.3V	3V3	Power Supply	Provides 3.3V power to RC522
	GND	GND	Ground	Common Ground
	RST	D3 (GPIO0)	Reset	Resets the RC522 module
	SDA / SS	D8 (GPIO15)	SPI Chip Select (CS)	Slave Select for SPI communication
	MOSI	D7 (GPIO13)	SPI Master Out	Data from ESP8266 to RC522
	MISO	D6 (GPIO12)	SPI Master In	Data from RC522 to ESP8266
	SCK	D5 (GPIO14)	SPI Clock	Clock signal for SPI communication
	IRQ	Not Connected	Interrupt	Not used in this project

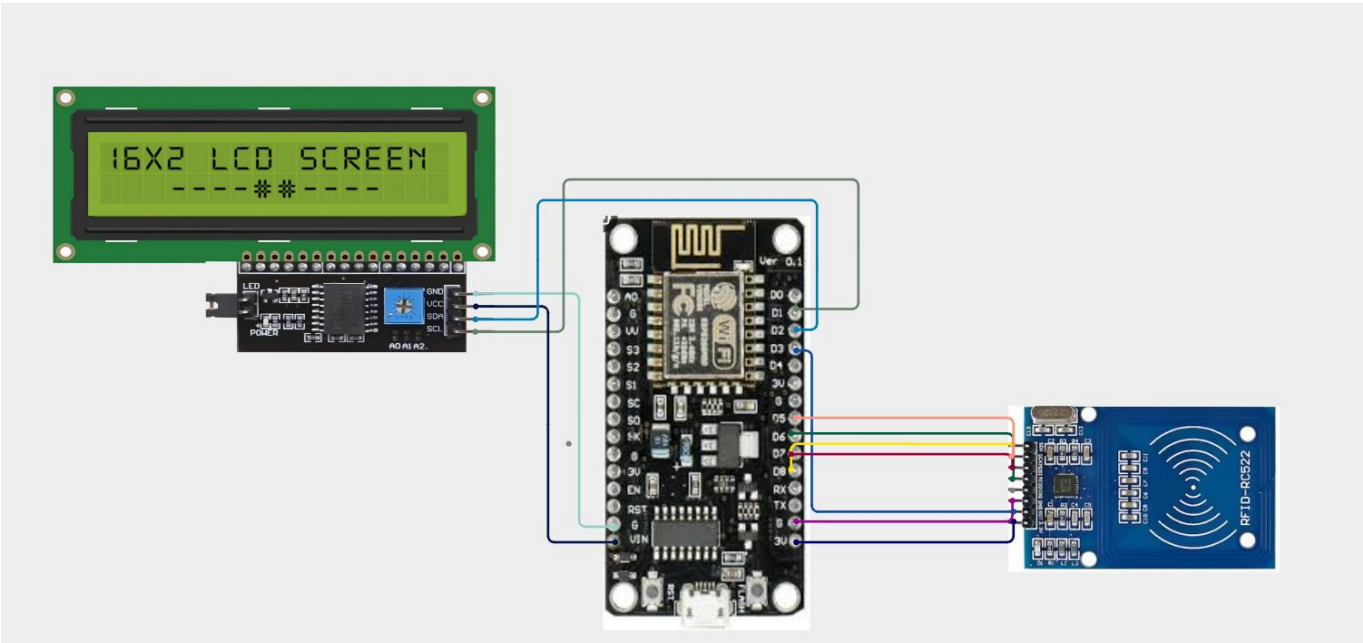
2. I2C LCD (16x2) TO ESP8266 (NODEMCU) CONNECTIONS

I2C LCD (16x2) 	LCD Pin	ESP8266 (NodeMCU) Pin	Function	Description
	VCC	VIN (5V)	Power Supply	Provides 5V power to LCD
	GND	GND	Ground	Common Ground
	SDA	D2 (GPIO4)	I2C Data	Serial Data Line (I2C)
	SCL	D1 (GPIO5)	I2C Clock	Serial Clock Line (I2C)

NOTES:

- RC522 works on 3.3V. Do NOT connect it to 5V.
- Make sure all GND connections are common.
- Use I2C LCD with backpack (PCF8574) for easy connection.
- All GPIO pins mentioned are for NodeMCU / ESP8266.

6.Circuit Design:



7.Algorithm:

1. Start the system.
2. Initialize the ESP8266 NodeMCU.
3. Initialize the RC522 RFID reader using SPI communication.
4. Initialize the 16×2 I2C LCD for displaying the system status.
5. Connect the ESP8266 to the Wi-Fi network.
6. Connect to Adafruit IO using MQTT communication.
7. Display “RFID SYSTEM – Scan Card...” on the LCD.
8. Wait for an RFID card/tag to be placed near the RC522 reader.
9. When a card is detected, read the unique identification number (UID) of the RFID card.
10. Compare the received UID with the registered UID values stored in the ESP8266 program.
11. If the UID matches a registered user:
 - Identify the corresponding person's name.
 - Display “WELCOME” and the person's name on the LCD.
 - Send the person's name to the rfid-name feed in Adafruit IO.
 - Send RFID status = 1 to the rfid-status feed.
12. If the UID does not match:
 - Identify the card as an unknown card.
 - Display “UNKNOWN CARD” and “ACCESS DENIED” on the LCD.
 - Send RFID status = 0 to the rfid-status feed.
13. Store/record the received data in Adafruit IO along with its timestamp.
14. After a short delay, clear the LCD.

15. Display “RFID SYSTEM – Scan Card...” again.

16. Wait for the next RFID card.

17. Repeat the process continuously.

8. Coding:

```
#include <ESP8266WiFi.h>

#include <SPI.h>

#include <MFRC522.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include "Adafruit_MQTT.h"

#include "Adafruit_MQTT_Client.h"

// ----- WiFi -----

#define WIFI_SSID  "YOUR_WIFI_NAME"

#define WIFI_PASSWORD "YOUR_WIFI_PASSWORD"

// ----- Adafruit IO -----

#define AIO_USERNAME "YOUR_ADAFRUIT_USERNAME"

#define AIO_KEY      "YOUR_ADAFRUIT_IO_KEY"

#define MQTT_SERVER  "io.adafruit.com"

#define MQTT_PORT    1883

// ----- RC522 -----

#define SS_PIN  D8

#define RST_PIN D3

MFRC522 rfid(SS_PIN, RST_PIN);
```

```
// ----- LCD -----  
  
LiquidCrystal_I2C lcd(0x27, 16, 2);  
  
// ----- MQTT -----  
  
WiFiClient client;  
  
Adafruit_MQTT_Client mqtt(  
  
    &client,  
  
    MQTT_SERVER,  
  
    MQTT_PORT,  
  
    AIO_USERNAME,  
  
    AIO_KEY  
  
);  
  
// ----- Adafruit IO Feeds -----  
  
Adafruit_MQTT_Publish nameFeed =  
  
    Adafruit_MQTT_Publish(  
  
        &mqtt,  
  
        AIO_USERNAME "/feeds/rfid-name"  
  
    );  
  
Adafruit_MQTT_Publish statusFeed =  
  
    Adafruit_MQTT_Publish(  
  
        &mqtt,  
  
        AIO_USERNAME "/feeds/rfid-status"  
  
    );  
  
// =====
```

```
// Wi-Fi Connection

// =====

void connectWiFi()

{

  Serial.print("Connecting to WiFi");

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  while (WiFi.status() != WL_CONNECTED)

  {

    delay(500);

    Serial.print(".");

  }

  Serial.println();

  Serial.println("WiFi Connected!");

}

// =====

// Adafruit IO Connection

// =====

void MQTT_connect()

{

  if (mqtt.connected())

  {

    return;

  }

}
```



```
Serial.print("Connecting to Adafruit IO...");

int8_t ret;

while ((ret = mqtt.connect()) != 0)

{

    Serial.println(mqtt.connectErrorString(ret));

    mqtt.disconnect();

    delay(5000);

}

Serial.println("Connected to Adafruit IO!");

}

// =====

// SETUP

// =====

void setup()

{

    Serial.begin(115200);

    // Start RFID

    SPI.begin();

    rfid.PCD_Init();

    // Start LCD

    Wire.begin(D2, D1);

    lcd.init();

    lcd.backlight();
```

```
lcd.setCursor(0, 0);

lcd.print("RFID SYSTEM");

lcd.setCursor(0, 1);

lcd.print("Connecting...");

// Connect Wi-Fi

connectWiFi();

// Connect Adafruit IO

MQTT_connect();

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("RFID SYSTEM");

lcd.setCursor(0, 1);

lcd.print("Scan Card...");

Serial.println("System Ready!");

}

// =====

// MAIN LOOP

// =====

void loop()

{

  MQTT_connect();

  // Check for RFID card

  if (!rfid.PICC_IsNewCardPresent())
```

```
{  
  
    return;  
  
}  
  
if (!rfid.PICC_ReadCardSerial())  
  
    {  
  
        return;  
  
    }  
  
// -----  
  
// Create UID string  
  
// -----  
  
String uid = "";  
  
for (byte i = 0; i < rfid.uid.size; i++)  
  
    {  
  
        if (rfid.uid.uidByte[i] < 0x10)  
  
            {  
  
                uid += "0";  
  
            }  
  
        uid += String(rfid.uid.uidByte[i], HEX);  
  
        if (i < rfid.uid.size - 1)  
  
            {  
  
                uid += ":";  
  
            }  
  
    }  
  
}
```

```
uid.toUpperCase();

Serial.println();

Serial.print("Card UID: ");

Serial.println(uid);

// -----

// Identify Person

// -----

String personName = "UNKNOWN";

// Replace these example UIDs

// with your actual RFID UIDs

if (uid == "93:4A:7B:21")

{

    personName = "SIVA";

}

else if (uid == "A3:B4:C5:D6")

{

    personName = "RAHUL";

}

else if (uid == "11:22:33:44")

{

    personName = "SANJAY";

}

// -----
```

```
// Registered Card

// -----

if (personName != "UNKNOWN")

{

    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("WELCOME");

    lcd.setCursor(0, 1);

    lcd.print(personName);

    Serial.print("Name: ");

    Serial.println(personName);

    // Send name

    if (nameFeed.publish(personName.c_str()))

    {

        Serial.println("Name sent to Adafruit IO");

    }

    else

    {

        Serial.println("Name sending failed");

    }

    // Status = 1

    if (statusFeed.publish(1))

    {
```

```
    Serial.println("RFID Status: 1");

}

}

// -----

// Unknown Card

// -----

else

{

    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("UNKNOWN CARD");

    lcd.setCursor(0, 1);

    lcd.print("ACCESS DENIED");

    Serial.println("Unknown RFID card");

    // Status = 0

    statusFeed.publish(0);

}

delay(3000);

// Return to scan screen

lcd.clear();

lcd.setCursor(0, 0);

lcd.print("RFID SYSTEM");

lcd.setCursor(0, 1);
```

```
lcd.print("Scan Card...");

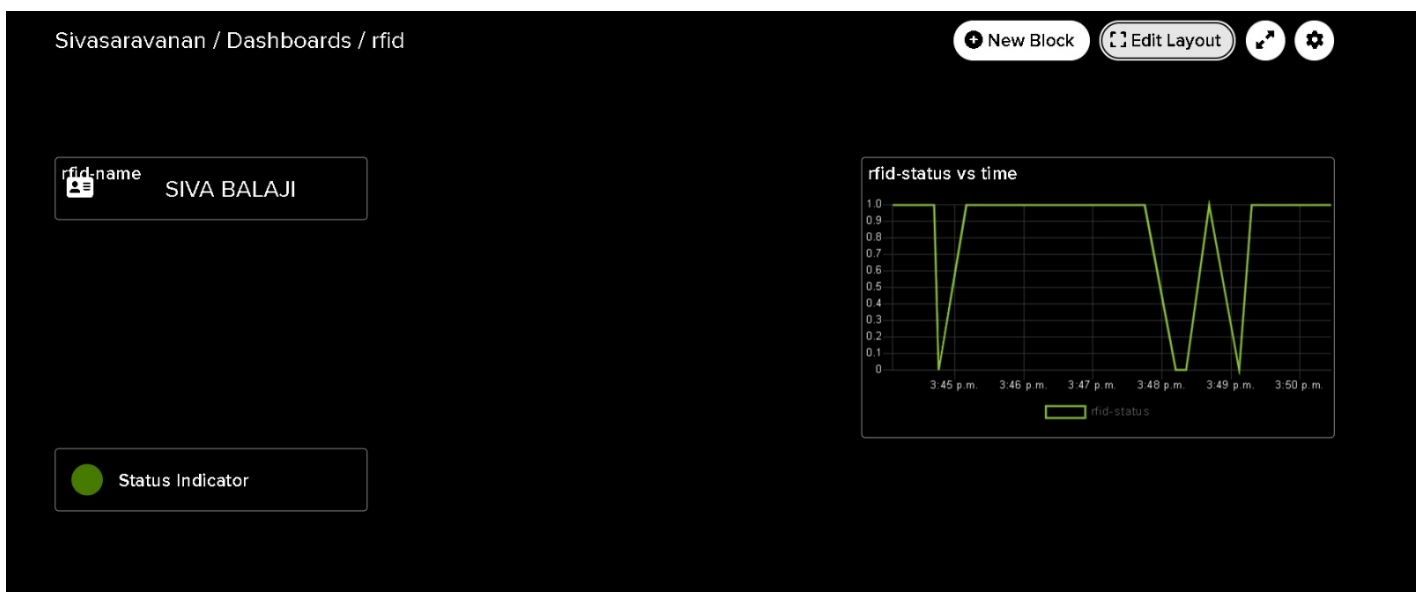
rfid.PICC_HaltA();

rfid.PCD_StopCrypto1();

delay(1000);

}
```

9. System Working :



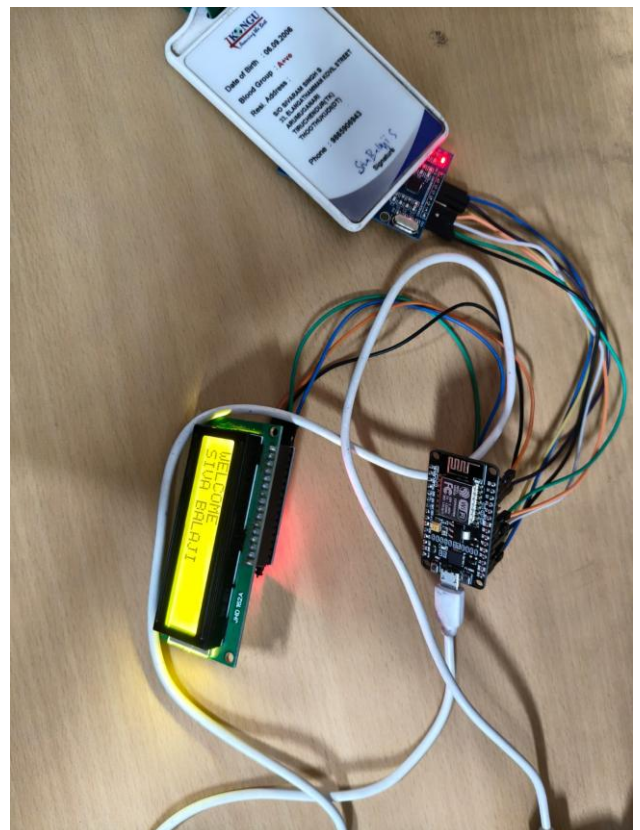
The system starts when power is supplied to the ESP8266. The ESP8266 initializes the RC522 RFID reader and I2C LCD and establishes a Wi-Fi connection.

When an RFID card is placed near the RC522 reader, the reader detects the card and reads its unique UID. The UID is transferred to the ESP8266 using the SPI communication protocol.

10. BILL OF MATERIAL:

BILL OF MATERIALS				
S. No.	Component / Part	Image	Quantity	Purpose / Description
1.	ESP8266 NodeMCU (Wi-Fi Module)		1	Main controller and Wi-Fi communication (Connects to Adafruit IO)
2.	RC522 RFID Reader Module		1	Reads RFID card/tag UID using 13.56 MHz
3.	RFID Card / Tag		2 or more	Used by users for identification
4.	16x2 I2C LCD Display (with I2C Backpack)		1	Displays status and user name
5.	Jumper Wires (Male to Male)		As required	For making electrical connections
6.	Breadboard		1	For prototyping and circuit connections
7.	Micro USB Cable		1	For power supply and programming
8.	5V Power Supply (via USB / Adapter)		1	Provides 5V power to the circuit
9.	Computer / Laptop		1	For coding, uploading and monitoring
10.	Internet / Wi-Fi		1	Required for connecting ESP8266 to Adafruit IO
11.	Adafruit IO Account		1	Cloud platform to store and visualize data

11. Prototype Implementation:



12. Results and Performance Analysis:

The developed RFID-Based Smart Identification and IoT Monitoring System was successfully implemented using an ESP8266 NodeMCU, RC522 RFID reader, 16×2 I2C LCD, and Adafruit IO cloud platform.

During testing, the RC522 successfully detected the RFID cards and read their unique identification numbers (UIDs). The ESP8266 compared the received UID with the registered UID stored in the program and identified the corresponding person.

For a registered RFID card, the person's name was successfully displayed on the LCD. The identified name was also transmitted to the Adafruit IO platform through Wi-Fi. The RFID status was recorded in the Adafruit IO feed, allowing the card scanning activity to be monitored remotely.

For an unregistered RFID card, the system displayed “UNKNOWN CARD” and “ACCESS DENIED” on the LCD. This demonstrated that the system could distinguish between registered and unregistered users.

The system provided fast contactless identification, reliable RFID reading, clear LCD display, and successful Wi-Fi-based data transmission. The Adafruit IO platform also enabled remote monitoring of the RFID identification data and status.

The prototype was tested with repeated card scanning and performed successfully under normal operating conditions. The system can be further developed for student attendance, employee attendance, library management, laboratory access control, office entry monitoring, and other IoT-based identification applications.

Overall, the developed system successfully achieved its objective of providing a low-cost, automated, contactless, and IoT-enabled RFID identification system.